

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, VII: THE SNAKE FACTORIZATION OF THE TRANSFER AND THE EVALUATED CHARACTERISTIC FORM

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ABSTRACT. Problem 5.1 of [V] asked for a characteristic formalism on the degeneracy tower satisfying three axioms: an arc-density term recovering $\mu = n(q+1)/q$ (A1), an additive Euler term $-\chi k$ with χ the rank of the degeneracy H^0 (A2), and reduction to the commutative characteristic ideal of [IV] on abelian sub-towers (A3). This paper solves the problem in evaluated form, by a single application of the snake lemma, and the mechanism turns out to be simpler than anything the series anticipated: because the level- $(k+1)$ Hecke operator factors through the tail norm, $A_{k+1} = \eta^* \tau_*$ [V, Prop. 1.2], it *vanishes identically* on $\ker \tau_*$, so the Laplacian restricted to the norm kernel is exactly qI and the “arc determinant” is exactly $q^{n_k(q-1)}$ — no nilpotency estimate, no unit ambiguity. The snake lemma applied to $0 \rightarrow \ker \nu \rightarrow \mathbb{Z}_p^{N_{k+1}} \rightarrow \mathbb{Z}_p^{N_k} \rightarrow 0$ with Laplacian verticals then yields, for every exactly intertwining, surjective, degree-preserving norm ν ,

$$\mathrm{ord}_p \mathrm{Tor}_{k+1} - \mathrm{ord}_p \mathrm{Tor}_k = \mathrm{ord}_p |\mathrm{coker}(\Delta_{k+1}|_{\ker \nu})| - \mathrm{ord}_p \mathrm{ind}_1(\nu),$$

where $\mathrm{ind}_1(\nu)$ is the factor by which ν scales the Eisenstein line. For $\nu = \tau_*$ this gives $n_k(q-1) - 1$: axiom (A1) with $\mathrm{Det}_{\mathrm{arc}} = q^{n_k(q-1)}$ and axiom (A2) with $\mathrm{Det}_{\mathrm{Euler}} = \mathrm{ind}_1(\tau_*) = q$, one factor per level, $\chi = 1$. For $\nu = \pi_*$ on an abelian voltage tower the same snake gives $\mathrm{ord}_\ell \prod_{\mathrm{new}} \zeta \det D(\zeta) - 1$: axiom (A3), the character mechanism of [II, §5] and [IV], verified to the integer $(\det(\Delta|_{\ker \pi}) = 48$ and 784 against the character products). One formula, two specializations, all three axioms now theorems. What is not delivered — a lift of this evaluated form to a single class in K_1 of a localized completed degeneracy algebra — is stated as the precise residue of Problem 5.1, and, consistently with [III] and [V], we decline to axiomatize a localization theory that does not exist.

STATUS OF THE RESULTS

All numbered statements are proved; the proofs use only results of Papers I–VI and the snake lemma. Every identity was additionally machine-verified exactly (Table 1): Iwahori transitions $1 \rightarrow 2$, $2 \rightarrow 3$ over K_4 , and abelian transitions $0 \rightarrow 1$, $1 \rightarrow 2$ of the voltage tower $(1, 1, 1)$, $\ell = 2$. Remark 3.1 states what remains open of [V, Prob. 5.1]; Remark 3.2 records two corrections to the task specification that prompted this paper. There are no external citations.

1. THE KERNEL OF THE NORM, AND THE VANISHING LEMMA

Setting as in [V]: Y_k the Iwahori path tower over a $(q+1)$ -regular base ($n_k = n(q+1)q^{k-1}$ for $k \geq 1$), τ_* the tail norm, A_k the level Hecke (adjacency) operator, $\Delta_k = qI - A_k$ for $k \geq 1$, sandpile torsion Tor_k .

Lemma 1.1 (Vanishing on the norm kernel). *For $k \geq 1$ let $K_k := \ker_{\mathbb{Z}_p}(\tau_*) \subseteq \mathbb{Z}_p^{n_{k+1}}$. Then:*

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Paper VII of the series; Papers I–VI are cited as [I]–[VI]. The verification script `iwahori_char.py` in the ancillary bundle reproduces every computational claim. This paper has no external bibliography: every input is a result of the series or textbook homological algebra.

- (i) K_k is a saturated direct summand of rank $n_{k+1} - n_k = n_k(q-1)$ (kernels of integer matrices are saturated);
- (ii) A_{k+1} vanishes on K_k : since $A_{k+1} = \eta^* \tau_* [V, \text{Prop. 1.2}]$, $A_{k+1}x = \eta^*(\tau_*x) = 0$ for $x \in K_k$; in particular K_k is invariant and

$$\Delta_{k+1}|_{K_k} = q \text{id}_{K_k}, \quad |\text{coker}(\Delta_{k+1}|_{K_k})| = q^{n_k(q-1)} \text{ exactly.}$$

Proof. (i) is the saturation principle of [II] (if $ka \in \ker$, $k \neq 0$, then $a \in \ker$); the rank is surjectivity of τ_* . (ii) is the displayed one-line computation; then $\Delta_{k+1} = qI - A_{k+1}$ restricts to $q \text{id}$, whose cokernel on a rank- $n_k(q-1)$ lattice has order exactly $q^{n_k(q-1)}$. Verified: the machine search for the nilpotency index of $A_{k+1}|_{K_k}$ returned $s = 1$ at both computed transitions, and $\det(\Delta_{k+1}|_{K_k}) = 2^{12}, 2^{24}$ on the nose. $\square$

Lemma 1.1(ii) is the structural surprise of this paper: the series expected, at best, a nilpotency estimate through the characteristic-polynomial ratio of [III, eq. (2.1)]; instead the incidence calculus of [V] kills the operator outright, because the Hecke operator of the upper level literally factors through the norm to the lower one.

2. THE SNAKE FACTORIZATION

Theorem 2.1 (Snake factorization of the transfer). *Let $\nu: \mathbb{Z}_p^{N_{k+1}} \rightarrow \mathbb{Z}_p^{N_k}$ be surjective and degree-preserving, with $\nu\Delta_{k+1} = \Delta_k\nu$, and suppose the Eisenstein lines satisfy $\nu(\mathbf{1}) = c\mathbf{1}$ for an integer $c =: \text{ind}_1(\nu) \neq 0$. Then $K := \ker \nu$ is a saturated invariant summand, the snake lemma applied to*

$$0 \longrightarrow K \longrightarrow \mathbb{Z}_p^{N_{k+1}} \xrightarrow{\nu} \mathbb{Z}_p^{N_k} \longrightarrow 0$$

with verticals $\Delta_{k+1}|_K, \Delta_{k+1}, \Delta_k$ yields the exact sequence

$$0 \rightarrow \mathbb{Z}_p \xrightarrow{c} \mathbb{Z}_p \xrightarrow{\delta} \text{coker}(\Delta_{k+1}|_K) \rightarrow \text{coker} \Delta_{k+1} \rightarrow \text{coker} \Delta_k \rightarrow 0,$$

and hence, whenever $\text{coker}(\Delta_{k+1}|_K)$ is finite,

$$(2.1) \quad \text{ord}_p \text{Tor}_{k+1} - \text{ord}_p \text{Tor}_k = \text{ord}_p |\text{coker}(\Delta_{k+1}|_K)| - \text{ord}_p c.$$

Proof. Invariance of K is the intertwining; saturation as in Lemma 1.1(i). The kernels of the verticals are: $\ker(\Delta_{k+1}|_K) = K \cap \mathbb{Z}_p\mathbf{1} = 0$ (the constants are not in $\ker \nu$ since $\nu\mathbf{1} = c\mathbf{1} \neq 0$), $\ker \Delta_{k+1} = \mathbb{Z}_p\mathbf{1}$, $\ker \Delta_k = \mathbb{Z}_p\mathbf{1}$, and the induced map between them is $\mathbf{1} \mapsto c\mathbf{1}$. The snake lemma gives the displayed sequence; the free ranks of the two cokernels match through ν (degree preservation, as in [IV, Lem. 1.3]), so taking orders of the finite parts yields (2.1). $\square$

Corollary 2.2 (Iwahori specialization: axioms (A1) and (A2)). *For $\nu = \tau_*$, $k \geq 1$: $c = q$ ($[V]$, $\tau_*\mathbf{1} = q\mathbf{1}$) and Lemma 1.1 give*

$$\text{ord}_p \text{Tor}_{k+1} - \text{ord}_p \text{Tor}_k = n_k(q-1) - 1,$$

recovering [III, Thm. 2.1] homologically. Summation yields the pseudo-Iwasawa law with $\mu = n(q+1)/q$ carried entirely by $\text{Det}_{\text{arc}} := \det(\Delta_{k+1}|_{\ker \tau_}) = q^{n_k(q-1)}$ (axiom A1), and the Euler term $-\chi k$ carried entirely by $\text{Det}_{\text{Euler}} := \text{ind}_1(\tau_*) = q$, one factor per level, with $\chi = 1 = \text{rk } \ker(\tau^* - \eta^*)$ by [V, Thm. 2.1] (axiom A2): the -1 is, finally and precisely, the Eisenstein line passing through the connecting map of the snake.*

Corollary 2.3 (Abelian specialization: axiom (A3)). *For an abelian $\mathbb{Z}/\ell^k$ voltage tower with covering pushforward $\nu = \pi_*$ (exactly intertwining by [IV, Lem. 1.2]): $c = \ell$, and under the block-character decomposition of [IV, Lem. 1.1] the kernel $\ker \pi_*$ is the sum of the new character blocks, on which Δ acts with determinant $\prod_{\text{new } \zeta} \det D(\zeta)$; hence (2.1) reads*

$$\text{ord}_\ell \text{Tor}_{k+1} - \text{ord}_\ell \text{Tor}_k = \text{ord}_\ell \prod_{\text{new } \zeta} \det D(\zeta) - 1,$$

which is precisely the character/Newton mechanism of [II, §5] and [IV, Cor. 2.3]. Verified exactly: $\det(\Delta|_{\ker \pi}) = 48 = \det D(-1)$ at $0 \rightarrow 1$ and $784 = |\det D(i) \det D(-i)|$ at $1 \rightarrow 2$, increments $4 - 1 = 3$ in both, matching [II, Table 2].

One snake, two kernels: on the Iwahori side the kernel is killed by the Hecke operator and contributes the pure arc power $q^{n_k(q-1)}$; on the abelian side the kernel is the new-character space and contributes the L -value block $\prod_{\text{new}} \det D(\zeta)$. The Eisenstein index c is q or ℓ respectively — one factor per level in either world, which is the whole content of $\lambda = -\chi$. The three axioms of [V, §5] are now theorems with these referents.

3. SCOPE, RESIDUE, CORRECTIONS

Remark 3.1 (The residue of Problem 5.1). Theorem 2.1 solves [V, Prob. 5.1] in *evaluated form*: the characteristic form is the level-wise pair $\text{Char}_k := (\det(\Delta_{k+1}|_{\ker \nu}), \text{ind}_1(\nu))$, and (A1)–(A3) hold for it as theorems. What is not constructed is a single element of K_1 of a localized completion of the degeneracy algebra — a Dieudonné-type determinant — whose evaluations reproduce Char_k . Defining such an object would require an Ore localization theory for the non-normal pro-ring of [V, Prop. 4.1], which does not exist; consistently with [III, Rem. 2.6] and [V, §5], we decline to axiomatize it, and we record the lift as the precise open residue. Everything quantitative that a lift could evaluate to is already proved here.

Remark 3.2 (Two corrections to the task specification). The specification prompting this paper asserted an exact sequence $0 \rightarrow X_{k+1} \xrightarrow{\partial} X_k \rightarrow 0$ with $\partial = \tau_* - \eta_*$; that map is far from injective ($\text{rk ker } \partial = n_{k+1} - n_k + 1$ by [V, Thm. 2.1(ii)]), and the exact sequence that carries the theory is the norm sequence of Theorem 2.1. It also placed the Eisenstein class in $K_0(\mathcal{H}_k)$; the Eisenstein projector $\mathbf{1}^t/N$ is not integral and does not lie in the algebra, and the class that exists is the module-level line $\mathbb{Z}_p \mathbf{1} = \ker \Delta$, entering the theory through the connecting map — which is where Corollary 2.2 puts it.

Remark 3.3 (Boundary level). At $k = 0$ the tail map fails to intertwine by exactly the degeneracy boundary ([V, Rem. 3.3]), $\ker S$ is not invariant, and Theorem 2.1 does not apply; the Eisenstein index there is $q + 1$, not q ($S\mathbf{1} = (q + 1)\mathbf{1}$, verified). The transfer $0 \rightarrow 1$ enters the law as the initial value, as everywhere in the series.

4. THE VERIFICATION BATTERY

transition	$\text{rk ker } \nu$	invariant	$A_{k+1} _{\ker}$	$\det(\Delta _{\ker \nu})$	$\text{ind}_1(\nu)$
Iwahori $1 \rightarrow 2$	12	yes	$= 0$ ($s=1$)	2^{12} exactly	2
Iwahori $2 \rightarrow 3$	24	yes	$= 0$ ($s=1$)	2^{24} exactly	2
abelian $0 \rightarrow 1$	4	yes	—	$48 = \det D(-1)$	2
abelian $1 \rightarrow 2$	8	yes	—	$784 = \det D(i) \det D(-i) $	2
boundary $0 \rightarrow 1$	—	no	—	—	$3 = q + 1$

TABLE 1. Exact verification (`iwahori_char.py`). Increments (2.1): $12 - 1 = 11$, $24 - 1 = 23$ (Iwahori, matching [II, Rem. 5.2]); $4 - 1 = 3$, $4 - 1 = 3$ (abelian 2-adic orders, matching [II, Table 2]). The vanishing $A_{k+1}|_{\ker \tau_*} = 0$ was found by the machine ($s = 1$) before its one-line proof was noticed.

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Unrefereed preprint. Every numbered statement is proved from Papers I–VI and the snake lemma; the battery of Table 1 confirms each ingredient exactly. No external sources are cited and none are load-bearing. Remark 3.1 states the open residue of [V, Prob. 5.1]; Remark 3.2 corrects the specification that prompted the paper.

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